

THABISO KOJOANA

Projects

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I hold a BSc in Chemistry and a BSc Honours in Chemistry from Wits, with a focus on computational chemistry during Honours. I am awaiting my final MSc Chemistry results. I began learning programming independently in 2022 and build applications with Java and Spring Boot.

Technology stack

Languages & web: Java, JavaScript, SQL, HTML5, CSS3; responsive layouts, DOM events and Fetch API.

Backend & databases: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, REST APIs, Bean Validation, PostgreSQL and Flyway.

Android: Android SDK, Room, WorkManager, CameraX, ML Kit and Material Components.

Testing & development: JUnit, Mockito, MockMvc, Spring Boot Test, Testcontainers, Git, Maven, Gradle and AI-assisted development with code review and testing.

Cloud & delivery: Docker; AWS EC2, RDS, ECR, S3 and CloudFront; Caddy HTTPS proxy.

Honours coursework: Computational chemistry, coding and data handling; supervised and unsupervised learning; neural-network concepts including convolutional (CNN) and recurrent (RNN) networks.

CAPSTONE PROJECT / PERSONAL PROJECT

Kasi-Till

Android point-of-sale system and owner portal

I built Kasi-Till to help South African spaza shop owners record sales and manage stock on an Android phone. The app stores data locally using Room, so sales can continue during internet outages and synchronise when the connection returns.

I developed Spring Boot REST APIs for products, staff access, sales and reporting, using PostgreSQL and Flyway. I added checks to prevent duplicate sales uploads and unauthorised access, implemented licence verification and deployed the Dockerised backend to AWS. I tested backend behaviour with JUnit, Mockito and controller tests.

[Watch demonstration](#) | [Open owner portal](#)

The portal requires an account. A walkthrough of the private source code is available on request.

Projects

Supporting projects | JavaScript, HTML and CSS

Five-Day Weather Studio

A weather simulator for six South African cities. Users can generate a five-day forecast and switch between Celsius and Fahrenheit. The cards show temperatures, rain chances and wind speeds. Changing units converts the values without generating a new forecast.

Demonstrates: arrays, state management, date handling, DOM rendering and responsive design. Values are simulated, not live weather data.

[Open demo](#) | [View source code](#)

Interactive CSS Robot

A robot drawn with CSS and controlled with JavaScript. Clicking its arm rotates it 90 degrees at the shoulder; clicking again lowers it. The arm also responds to keyboard controls, and the animation respects reduced-motion preferences.

Demonstrates: event handling, CSS transforms, animation state and keyboard interaction.

[Open demo](#) | [View source code](#)

Dream Car Builder

A car customiser with fields for make, model, year, body style, colour and price. Updating the form changes the car preview and its JavaScript object. The form checks for invalid entries, and a reset button restores the example.

Demonstrates: object modelling, form validation, safe text rendering and South African currency formatting.

[Open demo](#) | [View source code](#)

Name List Manager

A name list with controls to add, remove, sort and reverse entries. It rejects blank or duplicate names and displays input as text rather than HTML. Users can operate the controls with a keyboard and reset the list to its original examples.

Demonstrates: array operations, validation, DOM updates and accessible feedback.

[Open demo](#) | [View source code](#)

Train Animation Studio

An animated train with start, pause, speed, reverse, step and reset controls. The train turns around at each end of the track and pauses when the user leaves the tab. The scene adapts to different screen sizes.

Demonstrates: frame-based animation, elapsed-time movement, boundary handling, state management and keyboard-accessible controls.

[Open demo](#) | [View source code](#)

These web projects began as exercises from JavaScript For Kids For Dummies. I have added features and updated the interfaces to practise JavaScript, HTML and CSS.